ARCHITECTURE DEEP DIVE

SEMANTIC SEARCH

vs

KEYWORD SEARCH

vs

HYBRID

Why neither alone survives production and what actually works

THE PROBLEM

The Search Problem Nobody Talks About

Your search works perfectly in demos. It breaks in production. Here's why.

REAL SCENARIO

You build a product search for an e-commerce platform. You pick semantic search because it sounds modern and powerful.

WEEK 1 IN PRODUCTION

→ User searches **"Nike Air Max 270"** (exact product name)

→ Semantic search returns "Adidas Ultraboost" and "running shoes comfort"

→ The exact product exists in your catalog. The user can't find it.

→ Conversion drops. Users blame search. You blame embeddings.

THE REAL PROBLEM

→ Semantic search finds **meaning**. Keyword search finds **exact matches**.

→ Production users need both sometimes in the same query.

→ Neither alone is enough. This is why **Hybrid** exists.

KEYWORD SEARCH

How Keyword Search Works

Keyword search = find documents that contain the exact words you typed

THE ENGINE: BM25

- → Ranks documents based on term frequency and inverse document frequency
- → Words that appear often in a document but rarely in the corpus score higher
- → No understanding of meaning purely statistical matching

VISUAL: BM25 SCORING

```
Query: "python async programming"

Doc A: "Python async with asyncio"
→ Score: 8.4 ✓ HIGH

Doc B: "Async patterns in software"
→ Score: 3.1 ✕ LOW

Doc C: "Python web dev guide"
→ Score: 4.2 ✕ LOW
```

WHAT IT'S GOOD AT

- → Exact product names, model numbers, SKUs
- → Technical terms, error codes, proper nouns
- → Legal/medical terminology that must match precisely

WHAT IT MISSES

- → **Synonyms** "buy" vs "purchase"
- → **Intent** "for headache" should return paracetamol
- → **Multilingual** query in Hindi, document in English

SEMANTIC SEARCH

How Semantic Search Works

Find documents that mean what you're looking for, even if they use different words

THE ENGINE: DENSE VECTORS

- → Both query and documents are converted into high-dimensional vectors
- → Vectors closer together in space = more similar in meaning
- → Retrieved by cosine similarity or approximate nearest neighbor

VISUAL: VECTOR SPACE

```
"buy a car" ●——● "purchase auto"
      (very close) sim: 0.94

"buy a car"                    ●
                  "engine repair"
 ●
       (far apart) sim: 0.21
```

WHAT IT'S GOOD AT

- → **Intent understanding** "help me sleep" matches melatonin
- → **Synonym handling** "cheap" = "affordable"
- → **Multilingual** matches meaning across languages

WHAT IT MISSES

- → **Exact matching** "GPT-4o" vs "GPT-4"
- → **Rare terms** domain-specific terms
- → **Numeric precision** "rate 7.5%" vs "8.2%"

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COMPARISON

Head to Head: Where Each Breaks

Failure Mode Comparison Chart

QUERY TYPE	KEYWORD	SEMANTIC	WINNER
"Nike Air Max 270"	✓ HIGH	⚠ MED	**Keyword**
"shoes for bad knees"	✗ LOW	✓ HIGH	**Semantic**
"error code 404 fix"	✓ HIGH	⚠ MED	**Keyword**
"something to calm anxiety"	✗ LOW	✓ HIGH	**Semantic**
"iPhone 15 Pro Max"	✓ HIGH	⚠ MED	**Keyword**
"gift for my girlfriend"	✗ LOW	✓ HIGH	**Semantic**
"7.5% fixed rate mortgage"	✓ HIGH	✗ LOW	**Keyword**

THE PATTERN

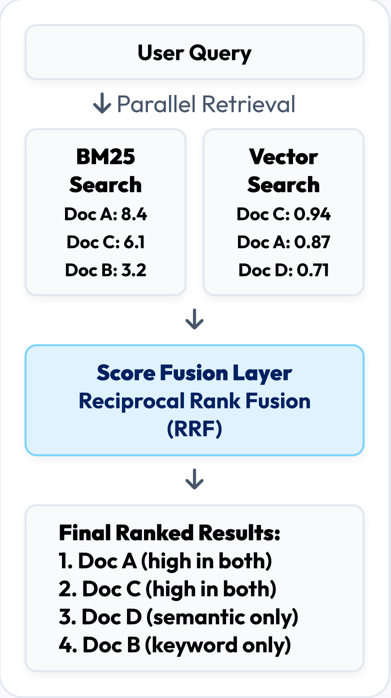
- → Exact terms, numbers, product names, codes Keyword wins
- → Intent, meaning, natural language, synonyms Semantic wins
- → Real users search both ways in the same session. **Pick one and you fail half your users.**

HYBRID SEARCH

How Hybrid Search Works

Run both BM25 and semantic search, fuse the results into one ranked list

THE PIPELINE



User Query

↓ Parallel Retrieval

BM25 Search
Doc A: 8.4
Doc C: 6.1
Doc B: 3.2

Vector Search
Doc C: 0.94
Doc A: 0.87
Doc D: 0.71

↓

Score Fusion Layer
Reciprocal Rank Fusion (RRF)

↓

Final Ranked Results:
1. Doc A (high in both)
2. Doc C (high in both)
3. Doc D (semantic only)
4. Doc B (keyword only)

THE FUSION ALGORITHM (RRF)

- **Reciprocal Rank Fusion**
- Score = 1/(rank_in_bm25 + k) + 1/(rank_in_semantic + k)
- Documents that rank high in BOTH systems get a massive combined boost
- Documents strong in only one system still surface but rank lower
- k = 60 is the standard constant

ALGORITHM

RRF Scoring Visualized

How Reciprocal Rank Fusion combines rankings

DOCUMENT	BM25 RANK	SEMANTIC RANK	RRF SCORE
Doc A	#1	#2	**0.031**
Doc C	#2	#1	**0.031**
Doc D	#8	#3	**0.024**
Doc B	#3	#12	**0.021**
Doc E	#15	#4	**0.019**

```
Formula: 1/(rank + 60) + 1/(rank + 60)
Doc A = 1/(1+60) + 1/(2+60) = 0.0164 + 0.0161 = 0.031
```

WHAT RRF DOES WELL

- → No need to normalize BM25 and cosine similarity scores to the same scale
- → Robust to outliers one bad rank in one system doesn't tank the whole result
- → Simple to implement, consistently outperforms weighted score blending

METRICS

Production Performance: The Numbers

Retrieval Accuracy by Query Type (NDCG@10)

QUERY TYPE	BM25	SEMANTIC	HYBRID
Exact term queries	0.82	0.61	**0.85** ▲
Intent queries	0.34	0.79	**0.83** ▲
Mixed queries	0.51	0.63	**0.81** ▲
Rare term queries	0.71	0.44	**0.74** ▲
Long queries	0.42	0.74	**0.79** ▲
Overall	0.56	0.64	**0.80** ▲

KEY TAKEAWAY

→ Hybrid doesn't just average the two it outperforms both on almost every query type.

→ On mixed queries (the majority of real production traffic): 0.81 vs 0.63 semantic vs 0.51 BM25.

→ The only time hybrid doesn't win cleanly is pure exact-term queries and even then it's comparable.

ENGINEERING

Implementing Hybrid Search

Implementation with Qdrant (production-grade vector DB)

```python
from qdrant_client import QdrantClient
client = QdrantClient("localhost", port=6333)

# 1. Index docs (dense + sparse)
client.upsert(
    collection_name="products",
    points=[{
        "id": doc_id, "payload": {"text": doc},
        "vector": {"dense": dense_vec, "sparse": sparse_vec}
    }]
)

# 2. Query both & fuse with RRF
results = client.query_points(
    collection_name="products",
    prefetch=[
        {"query": d_query, "using": "dense", "limit": 20},
        {"query": s_query, "using": "sparse", "limit": 20},
    ],
    query={"fusion": "rrf"}, limit=5
)
```

ALTERNATIVE IMPLEMENTATIONS

- → **Elasticsearch / OpenSearch** built-in hybrid search with RRF
- → **Weaviate** BM25 + vector search native hybrid
- → **LangChain EnsembleRetriever** combines any two retrievers with configurable weights

STRATEGY

When to Use Each in Production

The Decision Framework

What kind of queries does your system get?

Mostly exact terms
(codes, SKUs, names)
↓
Keyword (BM25)

Mostly natural language
(questions, intent)
↓
Semantic Search

Mix of both? (almost always)
↓
Hybrid + RRF

High accuracy needed fast?
Stop here

Best possible precision?
Add reranker on top

PRODUCTION RULES

- → Internal enterprise **Hybrid**
- → RAG for support bots **Hybrid**
- → Simple FAQ exact **BM25**
- → E-commerce search **Hybrid**
- → Creative / exploratory **Semantic**

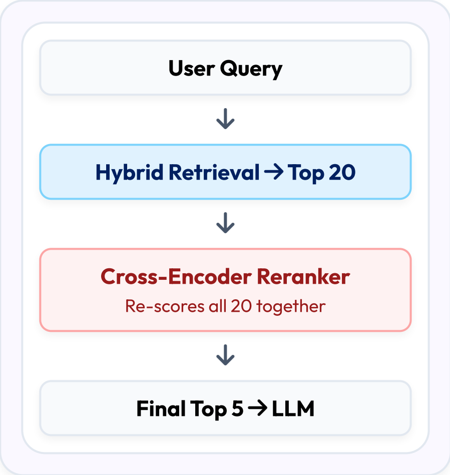
ADVANCED

Adding a Reranker on Top

Hybrid gives you candidates. A reranker picks the best one.

WHY RERANKING MATTERS

- → BM25 and semantic search use **bi-encoders** (query and document embedded separately)
- → A **cross-encoder reranker** reads query + document together much more accurate
- → Run it only on your top 20 hybrid results reorder into final top 5



User Query

↓

Hybrid Retrieval → Top 20

↓

Cross-Encoder Reranker

Re-scores all 20 together

↓

Final Top 5 → LLM

LATENCY TRADEOFF

- → Hybrid alone: **~50ms**
- → Hybrid + reranker: **~200ms**
- → For RAG and high-stakes retrieval: always worth it

```
from sentence_transformers import
CrossEncoder

reranker = CrossEncoder("ms-marco-MiniLM-
L-6-v2")
pairs = [[query, doc] for doc in
hybrid_top_20]
scores = reranker.predict(pairs)

# Sort by new score
reranked = sorted(zip(hybrid_top_20,
scores), key=lambda x: x[1],
reverse=True)
final_top_5 = [doc for doc, score in
reranked[:5]]
```

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★ SUMMARY

Cheatsheet

Full Comparison Table

CAPABILITY	KEYWORD	SEMANTIC	HYBRID
Exact match	✓ Best	⊖ Okay	✓ **Best**
Intent/meaning	✗ Poor	✓ Best	✓ **Best**
Synonyms	✗ Poor	✓ Best	✓ **Best**
Numbers/codes	✓ Best	✗ Poor	✓ **Best**
Cost	✓ Lowest	⊖ Medium	✗ **Higher**
Verdict	Baseline	Good start	**Production**

THE PRODUCTION STACK

Hybrid retrieval (BM25 + Dense + RRF)

+

Cross-encoder reranker on top 20

↓

Feed final top 5 into LLM

This is what Perplexity, Notion AI, and enterprise RAG systems run in production.

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